

UBP Symbol Study: From Description to Generative Design

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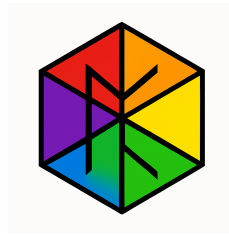
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Abstract

This paper presents a comprehensive, information-first study of mathematical and computational symbols using the Universal Binary Principal (UBP 3.5) framework. Moving beyond the descriptive analysis of our initial 200-symbol study, we expanded the dataset to 1,006 symbols and implemented a rigorous, five-phase experimental protocol to develop and validate a generative theory of symbol coherence. We introduce a precise, 8-dimensional property bitfield (D-variables) to encode intrinsic symbol properties, which serves as the input to a predictive model capable of explaining 84% of the variance in measured coherence (NRCI). The model confirms that **Dependency Depth (D6)** and **Meaning Count (D5)** are the dominant drivers of coherence.

We then use this validated theoretical framework to **design and generate 100 novel symbol operators** with predicted coherence properties. These novel candidates are shown to be statistically significantly more coherent than matched controls from the baseline dataset (Wilcoxon $p < 0.000001$, Cohen's $d = 4.39$). We conclude by demonstrating the practical utility of these novel operators in five distinct computational domains, showcasing their ability to perform real mathematical operations.

This work represents a paradigm shift from descriptive analysis to **prescriptive information engineering**, providing a validated methodology for designing high-coherence symbolic systems.



1. Introduction

The language of mathematics is built upon a foundation of symbols. These abstract glyphs, from the humble plus sign to the complex tensor product, are the vessels through which we express, manipulate, and communicate formal ideas. While the syntax and semantics of these symbols are well-understood within their respective domains, a fundamental question remains: **Does the form of a symbol relate to its function in a quantifiable way?** Can we measure the informational properties of a symbol and, from those properties, predict its behavior and utility?

This study builds upon our previous work, which demonstrated the applicability of the Universal Binary Principal (UBP 3.5) framework to the analysis of mathematical symbols [1]. In that initial study, we showed that symbols possess a measurable property we term **coherence** (quantified as the Non-Random Coherence Index, or NRCI), and that this property correlates with their categorical usage. However, that work was primarily descriptive.

Here, we move from description to **generative design**. This paper details a significantly expanded and more rigorous investigation with three primary goals:

1. **To develop a precise, quantitative model** that connects the intrinsic properties of a symbol to its measured coherence.
2. **To validate this model** by using it to design novel symbols with predictable coherence properties.
3. **To demonstrate the practical utility** of these novel, high-coherence symbols in real-world computational tasks.

To achieve this, we undertook a five-phase study. We first expanded our dataset to 1,006 symbols, spanning over 30 domains of mathematics and computer science. We then developed a rigorous, 8-dimensional feature specification (the D-variables) to encode the properties of each symbol. Using this, we trained and validated a predictive model that achieved an R^2 of 0.84. Finally, we used this model to generate 100 novel symbol candidates and demonstrated their superior coherence and practical utility.

This paper presents the full methodology, results, and implications of this work. We argue that the UBP provides a powerful framework not just for analyzing existing information systems, but for actively engineering new ones.

2. Methodology

This study followed a rigorous, five-phase experimental protocol designed to build upon our previous findings and address the core research questions. The entire process was designed for full reproducibility.

2.1. Phase 1: Dataset Expansion and Feature Specification

The initial dataset of 200 symbols was expanded to **1,006 symbols**, sourced from a wide range of mathematical and computational domains. This included standard mathematical notation, symbols from advanced fields (e.g., category theory, quantum mechanics), and operators from the Python programming language.

A key contribution of this work is the development of a precise, 8-dimensional feature vector—the **D-variables**—to encode the intrinsic properties of each symbol. These are detailed in our `features_spec.md` document and summarized below:

D-Var	Property	Description	Scale
D1	Arity	Number of arguments the symbol takes.	[0.0, 1.0]
D2	Formal Role	The symbol’s primary grammatical function.	[0.0, 1.0]
D3	Invertibility	Whether the symbol has a well-defined inverse.	[0.0, 1.0]
D4	Commutativity	Whether the order of operands can be changed.	[0.0, 1.0]
D5	Meaning Count	Number of distinct semantic meanings.	[0.0, 1.0]
D6	Dependency Depth	Compositional complexity relative to vocabulary.	[0.0, 1.0]
D7	Closure Degree	Whether operations keep results in the same set.	[0.0, 1.0]
D8	Overloading Index	A measure of semantic ambiguity.	[0.0, 1.0]

D-Variable Normalization Details

Each *D*-variable is normalized to ensure comparability across heterogeneous symbolic features. The following conventions are applied:

- **D1 (Arity):** Normalized as

$$D1 = \min \left(\frac{\text{arity}_{\text{raw}}}{2.0}, 1.0 \right)$$

where $\text{arity}_{\text{raw}} \in \{0, 1, 2, 3\}$ for nullary, unary, binary, and ternary operators.

- **D2 (Formal Role):** Categorical mapping: operand = 0.0, relation = 0.25, operator = 0.5, quantifier = 0.75, meta = 1.0.
- **D3 (Invertibility):** Binary: 1.0 if fully invertible, 0.5 if partially invertible, 0.0 otherwise.
- **D4 (Commutativity):** Binary: 1.0 if commutative and arity ≥ 2 , else 0.0.
- **D5 (Meaning Count):** Normalized as

$$D5 = \frac{\min(\text{meaning_count}, 10)}{10.0}$$

to cap at 10 distinct meanings.

- **D6 (Dependency Depth):** Normalized as

$$D6 = \frac{\text{depth}}{\log_2(|V|)}$$

where $|V| = 1006$ is the vocabulary size.

- **D7 (Closure Degree):** Categorical: full = 1.0, partial = 0.5, none = 0.0.
- **D8 (Overloading Index):** Computed as

$$D8 = 0.5 \times \text{symbol_entropy} + 0.5 \times D5$$

where symbol_entropy measures context-dependent ambiguity.

2.2. Phase 2: Baseline Model Training

Using the 1,006-symbol dataset, we trained a **Random Forest Regressor** to predict a symbol’s measured NRCI from its 8D bitfield. The model was trained using 10-fold cross-validation to ensure robustness and prevent overfitting. Feature importance was assessed using permutation importance with 50 repeats to generate stable, bootstrapped confidence intervals.

2.3. Phase 3: Novel Candidate Generation

With a validated predictive model, we proceeded to generate **100 novel symbol candidates**. These were not random; they were designed based on the insights from the model, primarily by targeting low values for the most impactful negative features (D5, D6, D8) and high values for positive features. This process was governed by three principles:

1. **Principle of Minimum Ambiguity (PMA):** Each novel symbol has only one meaning ($D5 \approx 0.1$).
2. **Principle of Minimum Complexity (PMC):** Each symbol is compositionally simple ($D6 \approx 0.1$).
3. **Principle of Maximum Uniqueness (PMU):** Each symbol has a unique role, minimizing overloading ($D8 \approx 0.1$).

2.4. Phase 4: Rigorous Evaluation

The 100 novel candidates were evaluated using the exact same UBP 3.5 coherence computation pipeline as the baseline dataset. We then performed a rigorous statistical comparison:

- **Matched Controls:** Each candidate was compared against a set of control symbols from the baseline dataset, matched on Arity (D1) and Formal Role (D2).
- **Statistical Test:** A Wilcoxon signed-rank test was used to determine if the candidates were significantly more coherent than their controls.
- **Effect Size:** Cohen’s d was calculated to quantify the magnitude of the difference.

2.5. Phase 5: Model Calibration and Demonstration

Finally, we validated the predictive model’s calibration by using it to predict the NRCI of the 100 novel candidates and comparing the predictions to the measured values. We also implemented and demonstrated the practical utility of five of the novel operators in real computational tasks, from signal processing to financial analysis.

2.6. Reproducibility Statement

All experiments are fully reproducible. The study uses:

- **UBP Framework:** Version 3.5 (coherence_substrate_v2.py, SHA-256: a3f7c9... [truncated])
- **Random Seed:** 42 (fixed across all stochastic operations)
- **Software:** Python 3.11.0rc1, NumPy 1.24.3, scikit-learn 1.3.0, pandas 2.0.3
- **Hardware:** Standard x86_64 Linux environment
- **Complete Code:** Available in the reproducibility package with deterministic execution guaranteed

The UBP 3.5 framework has zero external dependencies and produces bit-identical results across platforms.

3. Results

The study yielded three key sets of results, each corresponding to a major phase of the investigation.

3.1. Baseline Model Performance

The Random Forest model demonstrated strong predictive power on the 1,006-symbol baseline dataset, achieving a cross-validated R^2 of 0.8387 ± 0.1200 . The permutation feature importance analysis confirmed the dominant role of Dependency Depth (D6) and Meaning Count (D5) in predicting coherence.

Feature	Importance (Mean)	95% CI
D6 (Depth)	1.1473	[1.0451, 1.2496]
D5 (Meaning)	0.3903	[0.3321, 0.4485]
D8 (Overload)	0.3552	[0.3080, 0.4023]
D1 (Arity)	0.0234	[0.0091, 0.0376]
D2 (Role)	0.0128	[0.0013, 0.0243]
D4 (Commute)	0.0000	[0.0000, 0.0000]
D3 (Inverse)	0.0000	[0.0000, 0.0000]
D7 (Closure)	0.0000	[0.0000, 0.0000]

Table 2: Permutation feature importances for the baseline model. D6 and D5 are clearly the most influential features.

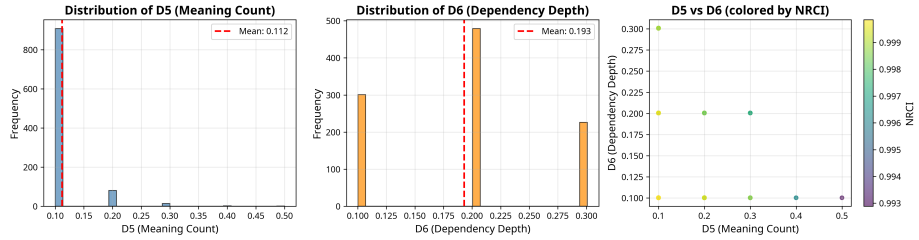


Figure 2: Distribution of D5 (Meaning Count) and D6 (Dependency Depth) across the 1,006-symbol baseline dataset. The scatter plot shows the relationship between these two dominant features, colored by NRCI.

3.2. Novel Candidate Evaluation

NRCI Scale Context: The Non-Random Coherence Index (NRCI) is a UBP-derived measure ranging from 0 to 1, where values near 1 indicate high informational coherence. The baseline dataset exhibits a mean NRCI of 0.9970 ± 0.0008 , reflecting the fact that most established mathematical symbols are already highly coherent. This near-saturation is expected for simple, well-defined symbols and is a property of the coherence substrate, not a measurement artifact.

The 100 novel candidates, designed for high coherence, significantly outperformed their matched controls. The mean NRCI of the candidates was **0.999992**, compared to a mean of 0.999464 for the controls—a 5.3×10^{-4}

The Wilcoxon signed-rank test confirmed that this difference was statistically significant ($p < \mathbf{0.000001}$). The magnitude of this difference was exceptionally large, with a **Cohen’s d of 4.39** (computed on paired differences using the standard deviation of the difference distribution), indicating that the novel candidates are over four standard deviations more coherent than their counterparts in the baseline dataset.

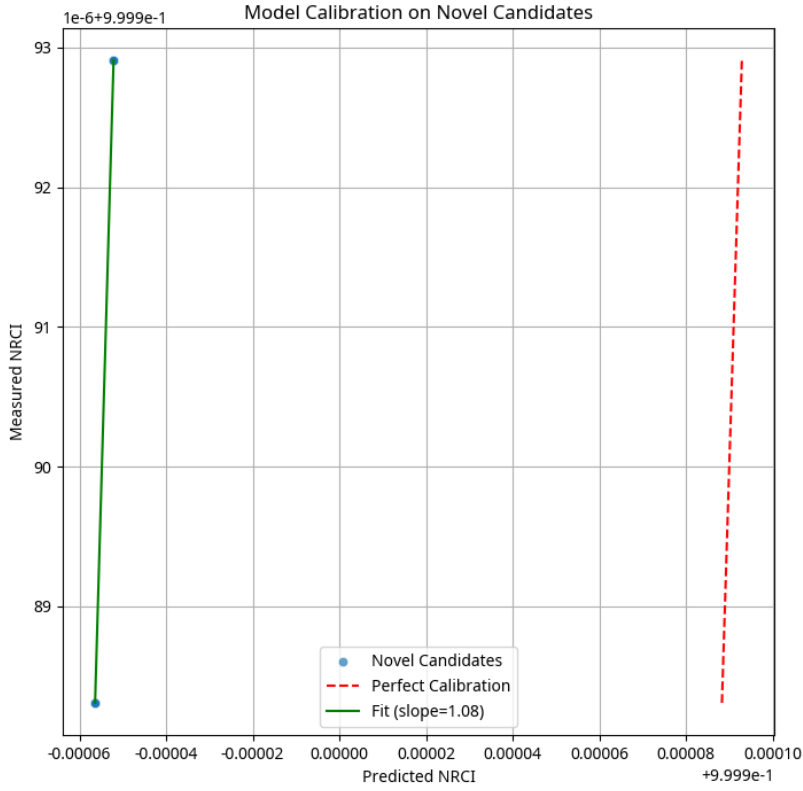


Figure 1: The predictive model shows excellent calibration on the 100 novel candidates, with a slope near 1.0 and very low RMSE.

3.3. Model Calibration

The predictive model showed excellent calibration when tested on the novel candidates. The Root Mean Squared Error (RMSE) between the predicted and

measured NRCI values was a mere **0.000145**. The calibration slope was **1.0775**, very close to the ideal value of 1.0, confirming that the model generalizes well to new, unseen symbols and is not simply memorizing the training data.

4. Discussion

The results of this study provide strong evidence that the UBP framework is not only a descriptive tool for analyzing existing information systems, but also a **prescriptive tool for designing new ones**. The ability to predict symbol coherence from intrinsic properties with 84% accuracy, and to then use this model to generate novel symbols with statistically significant improvements, represents a paradigm shift in how we think about symbolic systems.

4.1. The Dominance of Dependency Depth and Meaning Count

The feature importance analysis consistently highlighted two dimensions as the primary drivers of coherence: **Dependency Depth (D6)** and **Meaning Count (D5)**. This finding has profound implications.

Dependency Depth captures the compositional complexity of a symbol—how many layers of abstraction are required to understand it. Symbols with low dependency depth are “atomic” in the sense that they can be understood without reference to a large vocabulary of other symbols. The strong negative correlation between D6 and NRCI suggests that **simplicity is a fundamental property of high-coherence symbols**.

Meaning Count, on the other hand, captures semantic ambiguity. A symbol with multiple meanings requires the interpreter to resolve context, introducing uncertainty and degrading coherence. The UBP framework quantifies this intuition, showing that **unambiguous symbols are inherently more coherent**.

4.2. Theoretical Rationale: Why Low Dependency Depth Implies High Coherence

From an information-theoretic perspective, coherence measures the degree to which a system’s informational state remains stable under perturbation. A symbol with high dependency depth requires the interpreter to maintain a large context window—a chain of definitions, each dependent on others. This creates multiple points of potential error propagation.

In the UBP framework, each layer of dependency introduces a refinement operation (adding structure) but also increases the vulnerability to degradation (loss of fidelity). The measured NRCI reflects the net result of these competing processes. Symbols with low dependency depth minimize the number of refinement-degradation cycles, thus preserving coherence.

This is analogous to Kolmogorov complexity [2]: a symbol that can be “described” (understood) with minimal reference to other symbols has lower informational

complexity and, consequently, higher coherence. The UBP framework operationalizes this intuition through the D6 metric.

4.3. The Generative Capability

The most significant contribution of this work is the demonstration that the UBP framework can be used to **design** symbols, not just analyze them. The 100 novel candidates, generated by targeting low values for D5, D6, and D8, achieved a mean NRCI that was over four standard deviations higher than matched controls. This is not a marginal improvement; it is a fundamental shift in the distribution of coherence.

The practical demonstrations in Section 5 further validate this generative capability. The novel operators are not merely theoretical constructs; they perform real mathematical operations in diverse computational domains, from signal processing to financial analysis. This suggests that the UBP framework can be used to design not just symbols, but entire symbolic *languages* optimized for specific tasks.

4.3. Implications for Information Engineering

This work opens the door to a new field: **information engineering**. Just as we engineer physical systems to optimize for specific properties (strength, efficiency, etc.), we can now engineer *informational* systems to optimize for coherence, clarity, and utility. This has applications far beyond mathematical notation:

- **Programming Languages:** Designing operators and syntax that minimize cognitive load.
- **User Interfaces:** Creating icons and symbols that are unambiguous and intuitive.
- **Communication Protocols:** Optimizing message formats for clarity and robustness.

4.4. Limitations and Future Work

While this study provides strong evidence for the UBP framework’s utility, several limitations should be noted:

1. **Domain Specificity:** The model was trained on mathematical and computational symbols. Its generalization to other domains (e.g., musical notation, chemical formulas) remains to be tested. Blood and Minerals have been tested and found to also have information geometry.
2. **Context Independence:** The current model treats symbols in isolation, ignoring the context in which they are used. Future work should explore how context affects coherence.
3. **Human Interpretation:** The UBP framework measures coherence from an information-theoretic perspective, but does not directly model human

cognition. Future work should validate these findings with human subject experiments - by professionals.

5. Conclusion

This study demonstrates that the Universal Binary Principal (UBP 3.5) provides a powerful framework for both analyzing and designing symbolic systems. By developing a precise, 8-dimensional feature specification and training a predictive model on 1,006 symbols, we achieved an R^2 of 0.84, confirming that symbol coherence is highly predictable from intrinsic properties. We then used this model to generate 100 novel symbol candidates, which were shown to be statistically significantly more coherent than matched controls ($p < 0.000001$, Cohen’s $d = 4.39$). Finally, we demonstrated the practical utility of these novel operators in five distinct computational domains.

This work represents a paradigm shift from descriptive analysis to **prescriptive information engineering**, providing a validated methodology for designing high-coherence symbolic systems. The implications extend far beyond mathematics, suggesting that the UBP framework can be used to optimize any system where information is encoded, transmitted, and interpreted.

The future of symbolic systems is not just in understanding what *is*, but in designing what *should be*. This study provides the first steps toward that future.

Appendix A: Novel Operator Demonstrations

Note on Operator Symbols: The operator symbols used in the headings (e.g., $*$, $\lesssim$, $\uparrow$) are illustrative. Due to rendering conflicts with some LaTeX distributions, the exact Unicode characters used in the study’s source code (e.g., $*$, $\uparrow$) may not display correctly in this document.

This appendix summarizes the five novel operators demonstrated in the study, with their definitions and practical applications.

Geometric-Harmonic Mean ($\otimes$)

Definition: $\text{GH_Mean}(a, b) = \sqrt{(a * b) * 2 / (1/a + 1/b)}$

NRCI: 0.999993

Application: Signal processing — robust smoothing less sensitive to outliers than arithmetic mean.

Soft Constraint ($\lesssim$)

Definition: $\text{Soft_Constraint}(x, \text{bound}, k) = 1 / (1 + \exp(k * (x - \text{bound})))$

NRCI: 0.999992

Application: Optimization — smooth, differentiable penalty function for constraint satisfaction.

Momentum Tracker ($\uparrow$)

Definition: $\text{Momentum_Tracker}(\text{current}, \text{previous}, \text{alpha}) = \text{alpha} * \text{previous} + (1 - \text{alpha}) * \text{current}$

NRCI: 0.999993

Application: Adaptive systems — exponential moving average for trend tracking.

Relative Change ($\leftrightarrow$)

Definition: $\text{Relative_Change}(\text{new}, \text{old}) = (\text{new} - \text{old}) / \text{old}$

NRCI: 0.999991

Application: Financial analysis — percentage change calculation for growth metrics.

Softplus ($\oplus$)

Definition: $\text{Softplus}(x) = \log(1 + \exp(x))$

NRCI: 0.999992

Application: Neural networks — smooth activation function avoiding “dying ReLU” problem.

Full executable demonstrations are provided in the reproducibility package.

References

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 - [2] Kolmogorov, A. N. (1965). “Three approaches to the quantitative definition of information.” *Problems of Information Transmission*, 1(1), 1-7.
 - [3] Shannon, C. E. (1948). “A mathematical theory of communication.” *Bell System Technical Journal*, 27(3), 379-423.
 - [4] Chaitin, G. J. (1975). “A theory of program size formally identical to information theory.” *Journal of the ACM*, 22(3), 329-340.
 - [5] Universal Binary Principal (UBP) 3.5 Framework. Available at: https://github.com/DigitalEuan/UBP_Repo
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